

PROTECTED, IMPORTANT, OR BOTH?

Technical Methodology, Conservation Science, Ranking, Ecclesial
Storytelling, Stewardship Equivalents & Reproducible ArcPy

LSAP Integral Ecology Atlas | Response to the Cry of the Earth

Item	Value
Project	LSAP Integral Ecology Atlas
Map	Protected, Important, or Both?
Analytical scales	Dioceses; Episcopal Conferences; LSAP members at 20 km and 100 km
Environmental baseline	Esri ArcGIS Living Atlas Global Biodiversity and Conservation H3 hexagon summaries
Primary conservation signals	Formal protected-area coverage; Biodiversity Hotspots; SBTN conversion hotspots; Biodiversity Intactness Index
Final stewardship benchmark	1 acre supported per Catholic + 10 hectares supported per parish
Report date	3 October 2026
Status	Final analytical / publication methodology

This report documents the completed analysis. Symbology and final application presentation are intentionally outside scope.

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1. Executive summary

Protected, Important, or Both? is a four-scale conservation analysis in the LSAP Integral Ecology Atlas. It asks a deceptively simple question: where do formal protected-area coverage and globally recognized biodiversity importance coincide within Catholic ecclesiastical and Laudato Si' Action Platform geographies, where do they diverge, and what does that relationship mean for stewardship?

The map intentionally separates four concepts that are often collapsed into one another. Formal protection is represented by the protected-area share. Biodiversity importance is represented by Biodiversity Hotspot coverage. SBTN conversion hotspots are treated separately as an indicator of land-conversion pressure rather than as the definition of ecological importance. Biodiversity Intactness Index provides a modeled measure of ecological-community change relative to near-undisturbed reference conditions.

The map does not claim that formally protected land and Biodiversity Hotspot land occupy the exact same hectares within a diocese, Conference, or member buffer. The final context classes describe aggregate co-presence at the same analytical scale. When the project estimates how much Biodiversity Hotspot land may lie outside formal protection, it uses a mathematically defensible minimum and maximum range rather than inventing a precise spatial overlap that the summary data do not contain.

The final stories are explicitly ecclesial. Diocesan narratives begin with Catholics, parishes, and the bishop or sede vacante status. Episcopal Conference narratives begin with the Conference president, Catholic/parish scale, and Laudato Si' Action Platform participation. Member stories begin with the participating institution itself, then explain the 20 km local conservation context and 100 km regional relationship.

A final collective-stewardship thought experiment translates Church scale into an understandable area equivalent: one acre supported per Catholic plus 10 hectares supported per parish. It is explicitly hypothetical and does not imply land ownership, legal authority, or actual protected-area designation.

2. Final publication layers

Feature class	Map alias	Scale	Features	Final user fields
ProtectedImportantBoth_Dioceses_V2	Dioceses - Protection & Biodiversity Importance	Diocese	2,874	53
ProtectedImportantBoth_Conferences_V2	Episcopal Conferences - Protection & Biodiversity Importance	Episcopal Conference	122	73
ProtectedImportantBoth_Members_20km_V2	LSAP Members - 20 km Conservation Context	LSAP member / 20 km	2,739	65
ProtectedImportantBoth_Members_100km_V2	LSAP Members - 100 km Regional Conservation Context	LSAP member / 100 km	2,739	65

2.1 Final map group

The publication layers are placed in the ArcGIS Pro map group “Protected, Important, or Both?” in this order:

- LSAP Members - 20 km Conservation Context
- LSAP Members - 100 km Regional Conservation Context
- Dioceses - Protection & Biodiversity Importance
- Episcopal Conferences - Protection & Biodiversity Importance

3. Research question and analytical philosophy

The analytical question is not simply “where are protected areas?” or “where are biodiversity hotspots?” The atlas asks how those two systems relate to the Church geographies and LSAP communities that inhabit them.

1. Who is the Catholic community or institution represented here?
2. How much of its surrounding geography is formally protected?
3. How much falls within mapped Biodiversity Hotspot geography?
4. Does land-conversion pressure add additional urgency through SBTN conversion-hotspot coverage?
5. How intact is the ecological community according to the modeled Biodiversity Intactness Index?
6. How does this geography compare with peer dioceses, Conferences, or member landscapes?
7. For LSAP members, how does the conservation picture change from the immediate 20 km neighborhood to the wider 100 km region?
8. What kinds of stewardship and partnership are reasonable at this scale?

The guiding principle is relationship rather than scoring. “Protected” is not automatically equivalent to ecologically important; “important” is not automatically protected; and a lack of either broad indicator is not evidence that a place has no ecological value.

4. Source data and scientific provenance

4.1 Esri Global Biodiversity and Conservation Hexagons

The environmental foundation is Esri ArcGIS Living Atlas Global Biodiversity and Conservation Hexagons, summarized using nested H3 cells. Esri's July 2026 update reports 82 summary statistics, harmonized units,

improved treatment of partial pixels, and workflows explicitly designed for conservation planning, including identifying high-biodiversity areas, excluding areas already protected, and examining the remaining landscape.

The project used the Level-5 environmental summaries as the upstream science source. Esri describes Level-5 H3 cells as approximately 250 km² in the original 2025 release; the project's actual exported Level-5 source had a median HexArea of 257.22 km².

Sources: [Esri - Global Biodiversity and Conservation Hexagons...but better! \(2026\)](#) | [Esri - Introducing Global Biodiversity and Conservation Hexagons \(2025\)](#)

The ArcGIS item is distributed under CC BY 4.0, with separate CC BY-NC 4.0 terms for IUCN Red List-derived variables. Red List fields were retained in the working baseline but are not required for the final public story of this map.

ArcGIS item: [Global Hexagons for Biodiversity and Conservation](#)

4.2 Formal protection: Protected Planet / WDPA

The formal-protection indicator is based on the WDPA-derived protected-area percentage summarized in the environmental source. Protected Planet describes the World Database on Protected Areas as the most comprehensive global database of terrestrial and marine protected areas, maintained by UNEP-WCMC in collaboration with IUCN and data providers. The project preserves the source field meaning as formal protected-area context; it does not treat protection as proof of management effectiveness.

Source: [Protected Planet - Protected Areas \(WDPA\)](#)

4.3 Biodiversity Hotspots: Conservation International

Biodiversity Hotspots define the “ecologically important” dimension in this map. Conservation International describes hotspots as regions containing exceptional concentrations of endemic species that have also experienced severe habitat loss. To qualify, a region must contain at least 1,500 endemic vascular plant species and must have lost at least 70% of its original natural vegetation.

Source: [Conservation International - What Are Biodiversity Hotspots?](#)

4.4 SBTN conversion hotspots

SBTN is not folded into the definition of “important.” The source field is interpreted as land-conversion pressure. SBTN guidance describes conversion hotspots as places where pressures resulted in conversion of natural land classes to non-natural land classes between 2000 and 2020, with priority attention intended to prevent further conversion. SBTN identifies hotspots through rankings of percent and total conversion over 2000-2020 and 2010-2020.

Source: [Science Based Targets Network - Step 3 Land Technical Guidance](#)

4.5 Biodiversity Intactness Index

The Biodiversity Intactness Index (BII) is retained as supporting ecological context. The Natural History Museum describes BII as a modeled summary of ecological-community change under human pressure, with 100% representing a pristine reference and values above 90% associated with the proposed planetary-boundary range. In the public stories, the project avoids translating BII into a simplistic “health percentage.”

Source: [Natural History Museum - Biodiversity Intactness Index](#)

5. Four-scale baseline architecture

The analysis deliberately uses the project's established four biodiversity-summary layers as the authoritative analytical baselines, avoiding a second round of multi-million-row intersection processing for publication.

Scale	Baseline feature class	Records	Area field
Diocese	LSAP_Diocese_BiodiversitySummary	2,874	ANALYSIS_AREA
Episcopal Conference	LSAP_EpiscopalConferences_BiodiversitySummary	122	ANALYSIS_AREA
LSAP local	LSAP_MemberPoints_20km_BiodiversitySummary	2,739	BUFFER_AREA
LSAP regional	LSAP_MemberPoints_100km_BiodiversitySummary	2,739	BUFFER_AREA

The baseline layers already carry the weighted environmental summaries and the ecclesial or LSAP identity fields required for the final narratives. The member layers pair exactly 1:1 using BUFFER_ID; Account_Name is not used as a key because member names are not guaranteed unique.

6. Conservation indicators and scientific meaning

Publication field	Source meaning	Use in this map
PROTECTED_PCT	PCT_PROTECTED_AREA / WDPA-derived share	Formal protection dimension
IMPORTANT_PCT	PCT_BIODIVERSITY_HOTSPOT	Biodiversity-importance dimension
SBTN_PCT	PCT_SBTN_HOTSPOT	Separate land-conversion-pressure context
BIODIV_INTACT	WAVG_Biodiversity_Intactness_M	Modeled ecological-community condition
LANDSCAPE_KM2	ANALYSIS_AREA or BUFFER_AREA converted once	Explicit represented geography
PROTECTED_KM2	LANDSCAPE_KM2 x PROTECTED_PCT / 100	Protected-area context in km ²
IMPORTANT_KM2	LANDSCAPE_KM2 x IMPORTANT_PCT / 100	Biodiversity Hotspot context in km ²
SBTN_KM2	LANDSCAPE_KM2 x SBTN_PCT / 100	SBTN conversion-hotspot context in km ²

7. Level-5 H3 distribution audit and 10% context threshold

Before defining “substantive presence,” the project audited all 2,016,842 Level-5 hexagons. The objective was to avoid allowing a tiny boundary sliver to classify an entire approximately 257 km² H3 cell as protected or important.

Indicator	Positive cells (>0)	Cells >=10%	Share of positive cells retained at >=10%
WDPA formal protection	339,475	270,205	79.6%
Biodiversity Hotspot	433,416	428,486	98.9%
SBTN conversion hotspot	37,627	36,650	97.4%

Biodiversity Hotspot and SBTN fields were strongly binary-like: among positive cells, the median Biodiversity Hotspot share was 100%, and the median SBTN share was 100%. A 10% threshold therefore removes mostly marginal slivers while retaining nearly all genuinely positive hotspot cells. Formal protection is more continuously distributed, so the same 10% threshold intentionally distinguishes substantive protected-area presence from small protected-area fragments.

The 10% threshold is project-defined. It is not an official WDPA, Conservation International, SBTN, or Esri classification threshold.

Key threshold audit

LEVEL-5 SOURCE
Features: 2,016,842
Median HexArea: 257.22 km²
FORMAL PROTECTION - WDPA_Percent
Positive: 339,475 (16.83%)
>=10%: 270,205 (13.40% of all cells)
BIODIVERSITY HOTSPOT
Positive: 433,416 (21.49%)
>=10%: 428,486 (21.25% of all cells)
Positive-value median: 100%
SBTN CONVERSION HOTSPOT
Positive: 37,627 (1.87%)
>=10%: 36,650 (1.82% of all cells)
Positive-value median: 100%

8. Derived conservation-context classification

The final context classification is calculated from the continuous aggregate percentages in each baseline geography. It is a contextual classification, not a literal spatial-overlap calculation.

Substantive protection	PROTECTED_PCT >= 10 Formal protected-area coverage reaches the project-defined substantive-presence threshold.
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Substantive importance	IMPORTANT_PCT >= 10 Biodiversity Hotspot coverage reaches the project-defined substantive-presence threshold.
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Final class	Rule	Meaning
Protected + Important Landscape Context	Protection >=10% AND Hotspot >=10%	Both formal protection and biodiversity importance are significant within the same summarized geography.
Ecologically Important with Limited Formal Protection	Hotspot >=10% AND Protection <10%	Biodiversity-important geography is substantial, while formal protection is comparatively limited.
Formally Protected Landscape Context	Protection >=10% AND Hotspot <10%	Formal protection is substantial, while this particular hotspot indicator is less prominent.
Broader Landscape Context	Protection <10% AND Hotspot <10%	Neither broad indicator reaches the context threshold; this is not equivalent to “no ecological value.”

“Protected + Important” means aggregate co-presence at the scale of the feature. It does not establish that the protected hectares and Biodiversity Hotspot hectares are the same hectares.

8.1 Final context distributions

Scale	Protected + Important	Important / limited protection	Protected only context	Broader context
Dioceses	897 (31.2%)	639 (22.2%)	667 (23.2%)	671 (23.3%)
Episcopal Conferences	59 (48.4%)	11 (9.0%)	41 (33.6%)	11 (9.0%)
LSAP members - 20 km	716 (26.1%)	754 (27.5%)	252 (9.2%)	1,017 (37.1%)
LSAP members - 100 km	1,124 (41.0%)	511 (18.7%)	409 (14.9%)	695 (25.4%)

9. Ranking and percentile methodology

Formal protection, Biodiversity Hotspot coverage, and SBTN conversion-hotspot coverage are ranked independently within each analytical scale. Dioceses are compared only with dioceses, Conferences only with Conferences, 20 km member landscapes only with 20 km member landscapes, and 100 km member landscapes only with 100 km member landscapes.

Competition rank	$\text{Rank}(x) = 1 + \text{count}(\text{values} > x)$ Rank 1 is the highest value. Ties receive the same rank; subsequent ranks skip by the number of tied records.
Percentile	$\text{Percentile}(x) = 100 \times \text{count}(\text{values} \leq x) / N$ The percentage of same-scale records at or below the value.

The public integrated narrative uses ranking sparingly. Upper-quarter and upper-decile positions are translated into plain language, while exact rank and percentile remain available in dedicated fields and STORY_RANKING.

10. Area units and formula validation

Unit validation was treated as a required gate before area-based storytelling. The project confirmed that ANALYSIS_AREA and BUFFER_AREA are stored at square-meter scale, while the inherited conservation totals TOTAL_WDPA_Area, TOTAL_Biodiversity_Area, and TOTAL_SBTN_Hotspot_Area behave exactly as km².

Geography conversion	$\text{LANDSCAPE_KM2} = \text{raw area} / 1,000,000$ Applied exactly once. Raw area fields were not overwritten.
Protected area	$\text{PROTECTED_KM2} = \text{LANDSCAPE_KM2} \times \text{PROTECTED_PCT} / 100$ Matches inherited TOTAL_WDPA_Area.
Hotspot area	$\text{IMPORTANT_KM2} = \text{LANDSCAPE_KM2} \times \text{IMPORTANT_PCT} / 100$ Matches inherited TOTAL_Biodiversity_Area.
SBTN area	$\text{SBTN_KM2} = \text{LANDSCAPE_KM2} \times \text{SBTN_PCT} / 100$ Matches inherited TOTAL_SBTN_Hotspot_Area.

Area-unit QC

DIOCESES

Median ANALYSIS_AREA: 11,516,489,885.973 m²
Median LANDSCAPE_KM2: 11,516.490 km²
Derived/source ratio: 1.00000000

Inherited TOTAL_* median formula error: 0.00000000%

EPISCOPAL CONFERENCES

Median ANALYSIS_AREA: 280,242,973,406.825 m²

Median LANDSCAPE_KM2: 280,242.973 km²

Derived/source ratio: 1.00000000

Inherited TOTAL_* median formula error: 0.00000000%

LSAP MEMBERS - 20 KM

Median BUFFER_AREA: 1,256,636,981.225 m²

Median LANDSCAPE_KM2: 1,256.637 km²

Derived/source ratio: 1.00000000

LSAP MEMBERS - 100 KM

Median BUFFER_AREA: 22,556,872,817.134 m²

Median LANDSCAPE_KM2: 22,556.873 km²

Derived/source ratio: 1.00000000%

11. LSAP member 20 km to 100 km transition analysis

The 20 km and 100 km member derivatives pair exactly 1:1 using BUFFER_ID. Each publication layer carries both local and regional values so that either popup can tell the complete local-to-regional story.

Protection change	$PROT_CHANGE_PP = REGION_PROT_PCT - LOCAL_PROT_PCT$ Percentage-point change in formal protection share.
Importance change	$IMPORT_CHANGE_PP = REGION_IMPORT_PCT - LOCAL_IMPORT_PCT$ Percentage-point change in Biodiversity Hotspot share.
SBTN change	$SBTN_CHANGE_PP = REGION_SBTN_PCT - LOCAL_SBTN_PCT$ Percentage-point change in conversion-hotspot share.
BII change	$BII_CHANGE = REGION_BII - LOCAL_BII$ Change in modeled biodiversity intactness between scales.

Member transition QC

PAIRING

20 km unique BUFFER_ID: 2,739

100 km unique BUFFER_ID: 2,739

Duplicate IDs: 0

Only in one scale: 0

PASS - exact 1:1 pairing

MOST COMMON CONTEXT SHIFTS

661 Protected + Important at both scales

654 Broader Landscape at both scales

414 Important / limited protection -> Protected + Important

339 Important / limited protection at both scales

235 Broader Landscape -> Formally Protected

173 Formally Protected at both scales

104 Broader Landscape -> Important / limited protection

CHANGE DISTRIBUTIONS

Formal protection median change: +2.395 percentage points

Biodiversity Hotspot median change: 0.000 percentage points

SBTN median change: 0.000 percentage points

BII median change: +0.022

12. Ecclesial and LSAP storytelling architecture

The final integrated narrative is deliberately written for an intelligent general audience rather than for GIS specialists. Detailed technical interpretation remains available in separate STORY_SCIENCE, STORY_RANKING, and STORY_DATA_NOTE fields.

12.1 Diocese

- Identity: full jurisdiction name, Catholics, parishes, bishop/ordinary or sede vacante status.
- Scope: “Within the Diocese/Archdiocese/etc. of ...” appears before percentages.
- Conservation: explicit share “of the land” formally protected and within Biodiversity Hotspot geography.
- Comparison: upper-quarter/upper-decile ranking language where meaningful.
- Science: BII in accessible language; SBTN only surfaced when $\geq 10\%$.
- Collective stewardship: 1 acre supported per Catholic + 10 hectares supported per parish.
- Response: stewardship possibilities appropriate to the conservation class.

12.2 Episcopal Conference

- Identity: Conference president, Catholics, parishes.
- LSAP: enrolled participants and selected activity context.
- Scope: “Across the geography represented by [Conference]...”
- Conservation + ranking + BII + SBTN.
- Collective stewardship equivalent.
- Broad enabling response: knowledge sharing, diocesan capacity, partnerships, and connection to conservation expertise.
- Explore closer scale: directs readers to individual dioceses and 20 km / 100 km member landscapes.

12.3 LSAP members

- Identity: member name, sector, country.
- 20 km scope: “Within 20 km of [member]...”
- 100 km scope: “Across the 100 km regional landscape around [member]...”
- Conservation, ranking, BII, and SBTN context.
- Explicit local-to-regional shift in protection and hotspot coverage when materially different.
- Local or regional stewardship/partnership possibilities appropriate to scale.

13. Collective Catholic stewardship-equivalent framework

The stewardship metric is a thought experiment intended to make the social scale of the Church visible in spatial terms. It does not estimate land ownership, existing Catholic property, legal conservation authority, or actual protected-area designation.

Catholic stewardship equivalent	$STEW_ACRE_KM2 = Catholics \times 0.0040468564224$ One acre = 4,046.8564224 m ² = 0.0040468564224 km ² .
Parish stewardship equivalent	$STEW_PAR10HA_KM2 = Parishes \times 0.1$ Ten hectares = 0.1 km ² .
Combined stewardship equivalent	$STEW_COMBINED_KM2 = STEW_ACRE_KM2 + STEW_PAR10HA_KM2$ Combined people + institutional stewardship-equivalent area.

The public wording is intentionally “supported the stewardship of” rather than “protected” or “owned.” The final benchmark text is: “1 acre per Catholic + 10 hectares per parish.”

13.1 Brazil example

Component	Value
Catholics	95,925,851
Parishes	6,497
1 acre per Catholic equivalent	388,198 km ²
10 hectares per parish equivalent	650 km ²
Combined stewardship equivalent	388,848 km ²
Minimum guaranteed unprotected hotspot area	512,055 km ²
Maximum possible unprotected hotspot area	3,114,441 km ²
Combined equivalent relative to gap range	12%-76%

Final public interpretation: depending on how protected areas and Biodiversity Hotspots overlap on the ground, the combined stewardship equivalent represents at least about 12% of the hotspot land that could lie outside formal protection, and about 76% when compared with the minimum amount that must lie outside protection.

14. Mathematics of the unprotected-hotspot range

Because the publication baseline stores only aggregate areas, the exact protected-area/Biodiversity-Hotspot intersection is unknown. Set theory provides a valid range for the hotspot area outside protection.

Let T be total landscape area, I be Biodiversity Hotspot area, P be protected-area context, and O be the unknown intersection between I and P.

Possible overlap range	$\max(0, I + P - T) \leq O \leq \min(I, P)$ Fréchet bounds for the intersection of two subsets of the same total geography.
Unprotected hotspot	$U = I - O$ Hotspot area not overlapping formal protection.
Minimum possible U	$UNPROT_HOT_MIN_KM2 = \max(0, I - P)$ The amount of hotspot area that must lie outside protection, regardless of arrangement.
Maximum possible U	$UNPROT_HOT_MAX_KM2 = \min(I, T - P)$ The largest hotspot area that could lie outside protection.
Low stewardship share	$STEW_GAP_LOW_PCT = 100 \times \text{CombinedStewardship} / UNPROT_HOT_MAX_KM2$ Conservative comparison against the largest possible gap.

High stewardship share	$\text{STEW_GAP_HIGH_PCT} = 100 \times \text{CombinedStewardship} / \text{UNPROT_HOT_MIN_KM2}$ Comparison with the guaranteed minimum gap when that minimum is >0.
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The range is not statistical uncertainty. It is deterministic uncertainty caused by missing information about the internal spatial overlap of two aggregate area totals.

15. Public-language safeguards

- Every percentage explicitly refers to “the land” or to a named analysis geography.
- Member stories immediately state 20 km or 100 km scope rather than saying only “this landscape.”
- Diocese and Conference stories name the ecclesial geography before the environmental percentages.
- “Protected + Important” never claims literal spatial coincidence of the two mapped areas.
- BII is described as modeled ecological-community intactness, not a simple “health percentage.”
- SBTN is described as conversion-hotspot context, not as the definition of biodiversity importance.
- Stewardship equivalents use “supported the stewardship of” and explicitly disclaim ownership/control/legal authority.
- The broad-indicator caveat is moved to the end of Broader Landscape narratives rather than interrupting the story.

16. Quality-control record

The analysis used a small number of high-value scientific and publication checks. The outputs below are retained as the evidence base for the final publication derivatives.

16.1 Source and H3 audit

Retained QC output

AUTHORITATIVE LEVEL-5 SOURCE
GlobalHexagonsBiodiversityandConservationLevel15_Export
Features: 2,016,842
WDPA_Percent range: 0-100
Biodiversity_Area_Percent range: 0-100.000006
SBTN_Hotspot_Percent range: 0-100
Negative values: 0 for all three
>100.0001 values: 0 for all three

PAIRWISE CORRELATIONS

WDPA vs Biodiversity Hotspot:	0.136
WDPA vs SBTN:	-0.009
Biodiversity Hotspot vs SBTN:	0.015
WDPA vs BII:	0.071
Biodiversity Hotspot vs BII:	0.068
SBTN vs BII:	-0.054

16.2 Context classification QC

Retained QC output

DIOCESES

Protected + Important	897 (31.2%)
Ecologically Important / Limited Protection	639 (22.2%)
Formally Protected	667 (23.2%)
Broader Landscape	671 (23.3%)
Invalid derived percentages:	0
Missing ranks:	0

EPISCOPAL CONFERENCES

Protected + Important	59 (48.4%)
Ecologically Important / Limited Protection	11 (9.0%)
Formally Protected	41 (33.6%)
Broader Landscape	11 (9.0%)
LSAP MEMBERS - 20 KM	
Protected + Important	716 (26.1%)
Ecologically Important / Limited Protection	754 (27.5%)
Formally Protected	252 (9.2%)
Broader Landscape	1,017 (37.1%)
LSAP MEMBERS - 100 KM	
Protected + Important	1,124 (41.0%)
Ecologically Important / Limited Protection	511 (18.7%)
Formally Protected	409 (14.9%)
Broader Landscape	695 (25.4%)

16.3 Stewardship story QC

Retained QC output

FINAL STEWARDSHIP STORY QC

DIOCESES

Records updated: 2,874
Invalid negative km² values: 0
STORY_COLLECTIVE_STEWARDSHIP missing: 0
STORY_INTEGRATED_SUMMARY missing: 0

EPISCOPAL CONFERENCES

Records updated: 122
Invalid negative km² values: 0
STORY_COLLECTIVE_STEWARDSHIP missing: 0
STORY_INTEGRATED_SUMMARY missing: 0

Benchmark consistency after publication cleanup:

Incorrect benchmark text: 0 at both scales

16.4 Final publication cleanup

Retained QC output

DIOCESES

Before cleanup: 176 user fields
Removed: 123
Final: 53
Required fields missing: 0
Story fields: 9 | missing values: 0

EPISCOPAL CONFERENCES

Before: 321
Removed: 248
Final: 73
Required fields missing: 0
Story fields: 11 | missing values: 0

LSAP MEMBERS - 20 KM

Before: 150
Removed: 85
Final: 65
Required fields missing: 0
Story fields: 9 | missing values: 0

LSAP MEMBERS - 100 KM

Before: 150
Removed: 85
Final: 65
Required fields missing: 0

Story fields: 9 | missing values: 0

Final group:

Protected, Important, or Both?

Layer order:

LSAP Members - 20 km Conservation Context

LSAP Members - 100 km Regional Conservation Context

Dioceses - Protection & Biodiversity Importance

Episcopal Conferences - Protection & Biodiversity Importance

17. Final publication schemas

Layer	Final user fields	Story fields	Required-field QC
Dioceses	53	9	PASS
Episcopal Conferences	73	11	PASS
LSAP Members - 20 km	65	9	PASS
LSAP Members - 100 km	65	9	PASS

17.1 Core retained field families

- Verified geography and conservation areas in km².
- Protection, Biodiversity Hotspot, and SBTN percentages, classes, ranks, and percentiles.
- Biodiversity Intactness Index and project interpretation class.
- Diocesan Church identity and bishop/ordinary context.
- Episcopal Conference identity, leadership, Catholic/parish aggregates, and selected LSAP activity.
- Member identity, sector, country, and exact BUFFER_ID pairing.
- Member local/regional conservation values, ranks, changes, and context shift.
- Final stewardship-equivalent fields and unprotected-hotspot range fields for dioceses and Conferences.
- Accessible integrated stories plus detailed science/ranking/data-note fields.

18. Scientific limitations

- The four publication baselines are aggregate summaries. Exact internal overlap between protected land and Biodiversity Hotspot land is unknown.
- The 10% substantive-presence threshold is project-defined and intended as a sliver filter/context rule, not a source-provider threshold.
- Biodiversity Hotspots are one global prioritization framework and do not capture every locally important ecosystem or species assemblage.
- Formal protected-area coverage does not measure management effectiveness, governance quality, ecological condition, or connectivity.
- SBTN conversion hotspots describe historic land-conversion pressure and should not be treated as a generic biodiversity-importance index.
- BII is modeled and depends on reference conditions and underlying pressure-response modeling.
- Ecclesiastical geographies are organizational territories, not ecological boundaries.
- The stewardship-equivalent framework is illustrative. It is not a land-ownership estimate, a conservation pledge, a legal designation, or an estimate of additional protected area.
- The unprotected-hotspot range is mathematically exact given the aggregate inputs, but broad because it reflects missing spatial-overlap information.

- Ranks are comparison tools within a scale; they should not be interpreted as moral rankings of dioceses, Conferences, or LSAP members.

19. ArcGIS Online-ready metadata text

19.1 Summary

Protected, Important, or Both? explores how formal protected-area coverage and globally recognized biodiversity importance relate across Catholic dioceses, Episcopal Conferences, and the 20 km / 100 km landscapes around Laudato Si' Action Platform members. The analysis adds SBTN conversion-hotspot context, Biodiversity Intactness, scale-specific ranks, ecclesial identity, local-to-regional member transitions, and a collective Catholic stewardship thought experiment.

19.2 Description

This map is part of the LSAP Integral Ecology Atlas. Environmental indicators originate in Esri ArcGIS Living Atlas Global Biodiversity and Conservation H3 summaries and are analyzed through four established Catholic/LSAP summary layers. Formal protection is represented by protected-area coverage; ecological importance is represented by Biodiversity Hotspot coverage; SBTN conversion hotspots are treated separately as land-conversion-pressure context. A project-defined 10% threshold distinguishes substantive representation from marginal coverage. "Protected + Important" indicates that both aggregate shares meet the threshold within the same geography; it does not establish exact spatial overlap. The map also includes modeled Biodiversity Intactness, same-scale ranks and percentiles, LSAP member 20 km to 100 km transitions, Catholic leadership and participation context, and a hypothetical stewardship-equivalent framework of one acre supported per Catholic plus 10 hectares supported per parish.

19.3 Suggested tags

Protected Areas, WDPA, Protected Planet, Biodiversity Hotspots, Conservation International, SBTN, Conversion Hotspots, Biodiversity Intactness, Conservation Gap, H3, Esri Living Atlas, Laudato Si', Laudato Si' Action Platform, LSAP, Integral Ecology, Catholic Church, Dioceses, Episcopal Conferences, Parishes, Church Geography, Conservation, Habitat Connectivity, Stewardship, Response to the Cry of the Earth

19.4 Compact attribution

Ecclesiastical geography and Church statistics: LSAP Integral Ecology Atlas Catholic jurisdictional framework; Ecclesiastical Jurisdictional Boundaries – Pontifical University of Saint Thomas Aquinas, Catholic University of America; Holy See statistical publications; Catholic-Hierarchy.org. LSAP member/activity data: Laudato Si' Action Platform / Dicastery for Promoting Integral Human Development. Environmental context: Esri ArcGIS Living Atlas Global Biodiversity and Conservation Hexagons and underlying credited sources, including UNEP-WCMC / Protected Planet (WDPA), Conservation International Biodiversity Hotspots, Science Based Targets Network conversion hotspots, and Natural History Museum / PREDICTS Biodiversity Intactness Index.

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Jurisdictional boundaries Derived From:
Burhans, M., Bell, J., Burhans, D., Carmichael, R., Cheney, D., Deaton, M., Emge, T., Gerlt, B., Grayson, J., Herries, J., Keegan, H., Skinner, A., Smith, M., Sousa, C., Trubetskoy, S. (2016). *Diocesan Boundaries of the Catholic Church* (Version 1.0) [Data set]. GoodLands Inc., Environmental Systems Research Institute, Inc. (Esri).

Project sources: LSAP_Diocese_BiodiversitySummary; LSAP_EpiscopalConferences_BiodiversitySummary; LSAP_MemberPoints_20km_BiodiversitySummary; LSAP_MemberPoints_100km_BiodiversitySummary; LSAP Catholic jurisdictional framework; Laudato Si' Action Platform member/activity extract; Holy See statistical publications; Catholic-Hierarchy.org.

Appendix A. Reproducible ArcPy science code

The following code consolidates the scientific and reproducibility-critical logic used in the completed workflow. The executed project notebook contains additional text-generation and publication-cleanup utilities.

A.1 Core conservation context and ranks

```
from collections import Counter

SUBSTANTIVE_THRESHOLD = 10.0

def clamp_pct(value):
    if value is None:
        return None
    return max(0.0, min(100.0, float(value)))

def substantive(value):
    return value is not None and float(value) >= SUBSTANTIVE_THRESHOLD

def conservation_context(protected_pct, important_pct):
    protected = substantive(protected_pct)
    important = substantive(important_pct)

    if protected and important:
        return "Protected + Important Landscape Context"
    if important and not protected:
        return "Ecologically Important with Limited Formal Protection"
    if protected and not important:
        return "Formally Protected Landscape Context"
    return "Broader Landscape Context"

def make_rank_lookup(values):
    clean = [float(v) for v in values if v is not None]
    n = len(clean)
    counts = Counter(clean)

    rank_lookup = {}
    greater = 0
    for value in sorted(counts, reverse=True):
        rank_lookup[value] = 1 + greater
        greater += counts[value]

    percentile_lookup = {}
    cumulative = 0
    for value in sorted(counts):
        cumulative += counts[value]
        percentile_lookup[value] = cumulative / n * 100.0

    return {
        value: (rank_lookup[value], percentile_lookup[value])
        for value in counts
    }
```

A.2 Verified area calculations

```
# Raw ANALYSIS_AREA / BUFFER_AREA fields are square meters.
landscape_km2 = raw_area_m2 / 1_000_000.0

protected_pct = clamp_pct(PCT_PROTECTED_AREA)
important_pct = clamp_pct(PCT_BIODIVERSITY_HOTSPOT)
sbtn_pct = clamp_pct(PCT_SBTN_HOTSPOT)

protected_km2 = landscape_km2 * protected_pct / 100.0
important_km2 = landscape_km2 * important_pct / 100.0
sbtn_km2 = landscape_km2 * sbtn_pct / 100.0

# QC confirmed these reproduce inherited TOTAL_* fields in km².
```

A.3 Member local-to-regional transition

```
# BUFFER_ID is the authoritative 1:1 pairing key.

PROT_CHANGE_PP = REGION_PROT_PCT - LOCAL_PROT_PCT
IMPORT_CHANGE_PP = REGION_IMPORT_PCT - LOCAL_IMPORT_PCT
SBTN_CHANGE_PP = REGION_SBTN_PCT - LOCAL_SBTN_PCT
BII_CHANGE = REGION_BII - LOCAL_BII

if LOCAL_CONSERV == REGION_CONSERV:
    CONSERV_SHIFT = f"{LOCAL_CONSERV} at both scales"
else:
    CONSERV_SHIFT = f"{LOCAL_CONSERV} -> {REGION_CONSERV}"
```

A.4 Unprotected-hotspot range

```
def unprotected_hotspot_range(
    landscape_km2,
    important_km2,
    protected_km2
):
    T = float(landscape_km2)
    I = float(important_km2)
    P = float(protected_km2)

    minimum = max(0.0, I - P)
    maximum = min(I, max(0.0, T - P))

    return minimum, maximum
```

A.5 Collective Catholic stewardship equivalents

```
ACRE_KM2 = 0.0040468564224
TEN_HECTARES_KM2 = 0.1

STEW_ACRE_KM2 = Catholics * ACRE_KM2
STEW_PAR10HA_KM2 = Parishes * TEN_HECTARES_KM2
STEW_COMBINED_KM2 = STEW_ACRE_KM2 + STEW_PAR10HA_KM2

UNPROT_HOT_MIN_KM2, UNPROT_HOT_MAX_KM2 = \
    unprotected_hotspot_range(
        LANDSCAPE_KM2,
        IMPORTANT_KM2,
        PROTECTED_KM2
    )

STEW_GAP_LOW_PCT = (
    STEW_COMBINED_KM2 / UNPROT_HOT_MAX_KM2 * 100.0
    if UNPROT_HOT_MAX_KM2 > 0 else None
)

STEW_GAP_HIGH_PCT = (
    STEW_COMBINED_KM2 / UNPROT_HOT_MIN_KM2 * 100.0
    if UNPROT_HOT_MIN_KM2 > 0 else None
)
```

A.6 Accessible scope-specific story structure

```
# Diocese scope
scope = f"Within the {DioType} of {DioName}"

# Member local scope
scope = f"Within 20 km of {Account_Name}"

# Member regional scope
scope = (
    f"Across the 100 km regional landscape around {Account_Name}"
)

# Conference scope
scope = (
```

```

        f"Across the geography represented by the {ConferenceName}"
    )

# Public conservation sentence
story = (
    f"{scope}, about {PROTECTED_PCT:.0f}% of the land is formally "
    f"protected, while about {IMPORTANT_PCT:.0f}% of the land falls "
    f"within mapped biodiversity-hotspot geography."
)

```

A.7 Publication cleanup pattern

```

# Final derivatives retain only publication-relevant fields.
existing = {f.name for f in arcpy.ListFields(fc)}
publication_keep = KEEP_FIELDS & existing

delete_fields = [
    f.name
    for f in arcpy.ListFields(fc)
    if not f.required and f.name not in publication_keep
]

if delete_fields:
    arcpy.management.DeleteField(fc, delete_fields)

# Required-field, story-completeness, and stewardship-unit QC follow.

```

Appendix B. Retained QC logs

Retained QC output

FINAL PUBLICATION STATUS

DIOCESES

Features: 2,874
Final user fields: 53
Required fields missing: 0
Story fields: 9
Missing story values: 0

EPISCOPAL CONFERENCES

Features: 122
Final user fields: 73
Required fields missing: 0
Story fields: 11
Missing story values: 0

LSAP MEMBERS - 20 KM

Features: 2,739
Final user fields: 65
Required fields missing: 0
Story fields: 9
Missing story values: 0

LSAP MEMBERS - 100 KM

Features: 2,739
Final user fields: 65
Required fields missing: 0
Story fields: 9
Missing story values: 0

ANALYTICAL STATUS

Area units verified before publication.
Aggregate context threshold = 10%.
Protection / hotspot / SBTN ranks complete.
Exact member pairing by BUFFER_ID.
Unprotected-hotspot range derived with set-theoretic bounds.
Final stewardship benchmark = 1 acre/Catholic + 10 ha/parish.
No negative stewardship km² values.
Benchmark text inconsistencies = 0.
Integrated public stories complete at all scales.

PUBLICATION GROUP

Protected, Important, or Both?